PCEOT502 Power Electronics

Course Outcomes

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the operation of modern power semiconductor devices, its characteristics and Select suitable gate driver circuits & heatsinks	**K5**
CO2	Understand the features of phase-controlled rectifiers, AC voltage Controllers & Switching Regulators and Analyse the operation	**K4**
CO3	Understand the features of different types of switch mode DC-AC Inverters and Analyse the operation	**K3**
CO4	Understand the need for improved efficiency, improved reliability, improved load & source waveforms and improved utility interface	**K3**
CO5	Understand the features of adjustable speed drives and Analyse the basic drive schemes for DC motors and Induction Motors	**K4**

Note: *K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Power Electronics- Converters, Applications and Design, 3ed(Indian Adaptation) by Mohan, Undeland, Robbins, Wiley India, 2022	Ned Mohan, Undeland, Robbins	Wiley-India	2022
2	Power Electronics- Principles and Applications	Joseph Vithayathil	Tata Mcgraw Hill	2010
3	Power Electronics	Cyril W Lander	McGrawHill	1993
4	Power Electronics – Circuits, Devices and Applications	Muhammad H. Rashid	Pearson Education	2014
5	Power Electronics	D.W. Hart	McGrawHill	2010
6	Power Electronics – Essentials & Applications	L. Umanand	Wiley-India	2009
7	Fundamentals of Electric Drives	G K Dubey	Narosa	2001

Module 1- Power Switches

Module 2- AC- DC and DC-DC Converters

Module 3- AC to AC and DC-AC Converters

Module 4- DC and AC Drives

Module 1

Role of Power Electronics, Motivation, Objectives and Challenges, Power Electronics Vs Linear Electronics, Ideal and real switches- Static and dynamic Performance – Power losses- Temperature rise- Thermal Analogy- Use of Heat sinks- Need for high efficiency, small size, high reliability and low cost- Overview of Applications [2 Hrs]

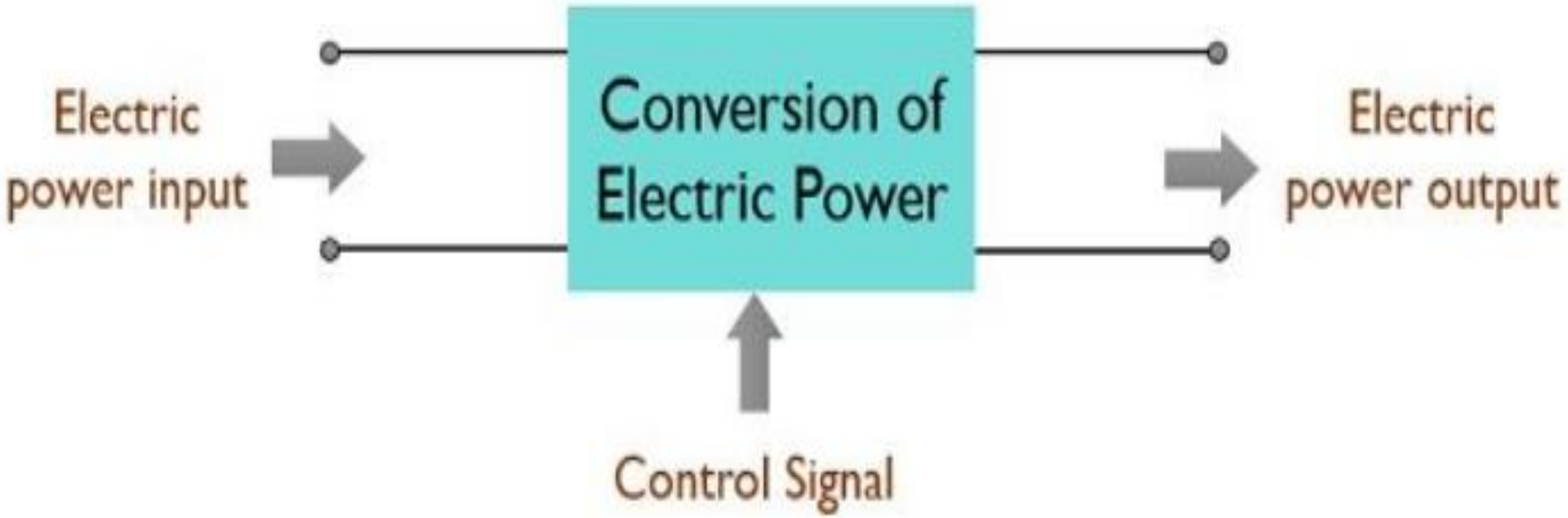
Power Electronics

Branch of electrical engineering which deals with the behavior and operation of high voltage and current devices used to control and deliver electrical power.

Involves the study of :-

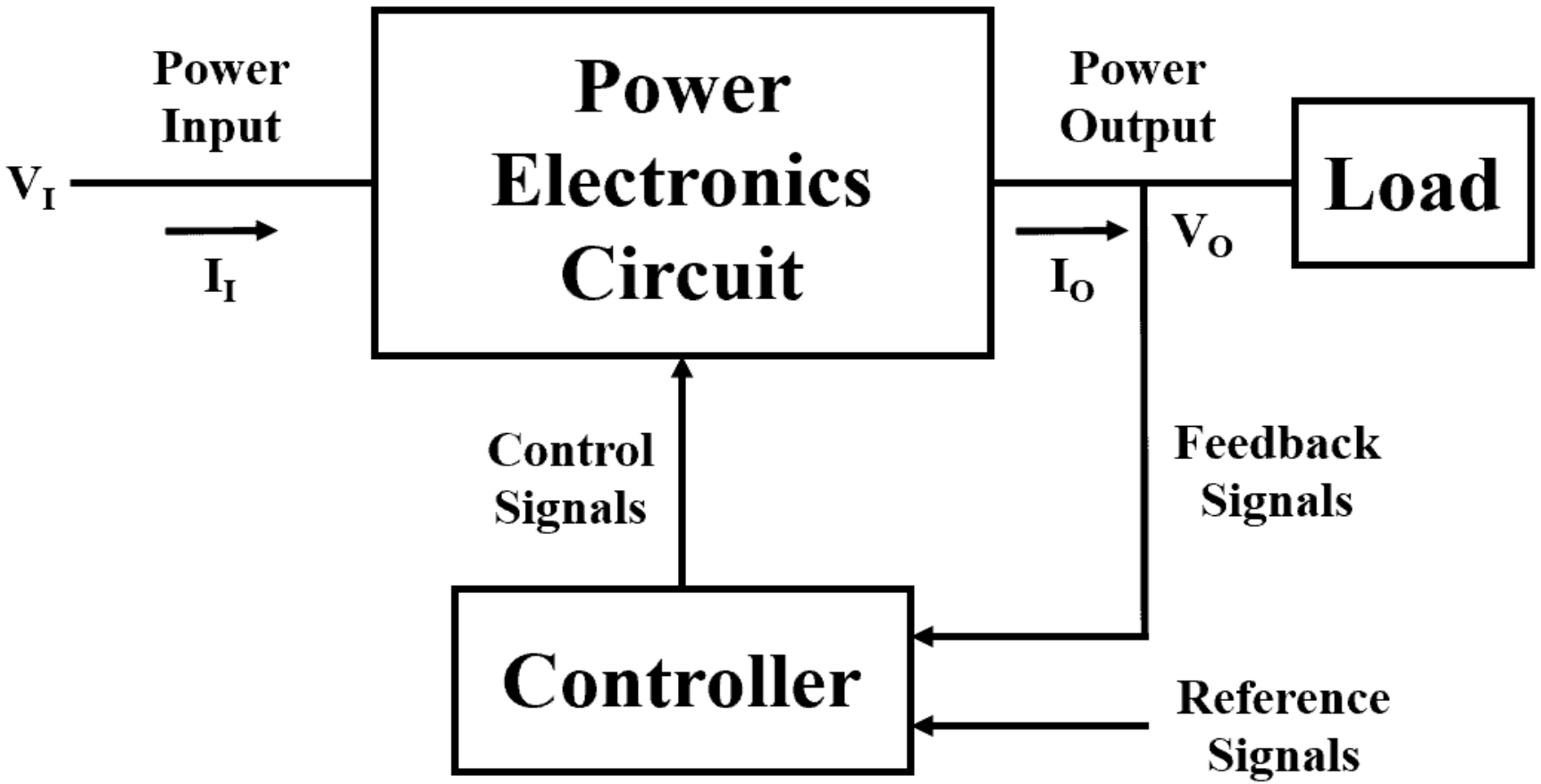
- Power semiconductor devices –their physics, chara, drive requirements and their protection
- Power converter topologies
- Control strategies of the converters
- Digital, analog and microelectronics involved
- Magnetic and capacitive storage elements
- Static and rotating electrical drives
- Analysis of the generated waveforms
- EM and RF interference

Power Electronic System

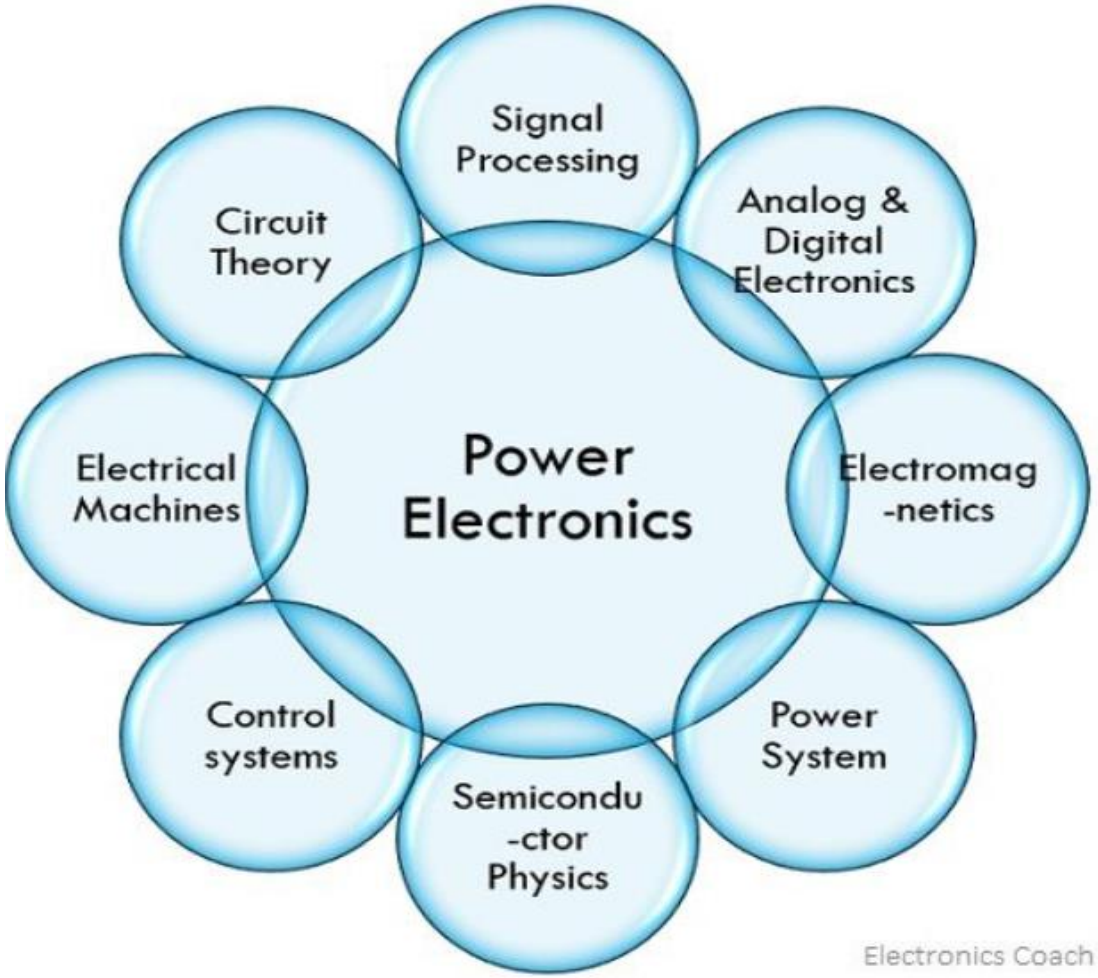


Representation of Electric Power Conversion

Power Electronic System (Contd…)



Power Electronics



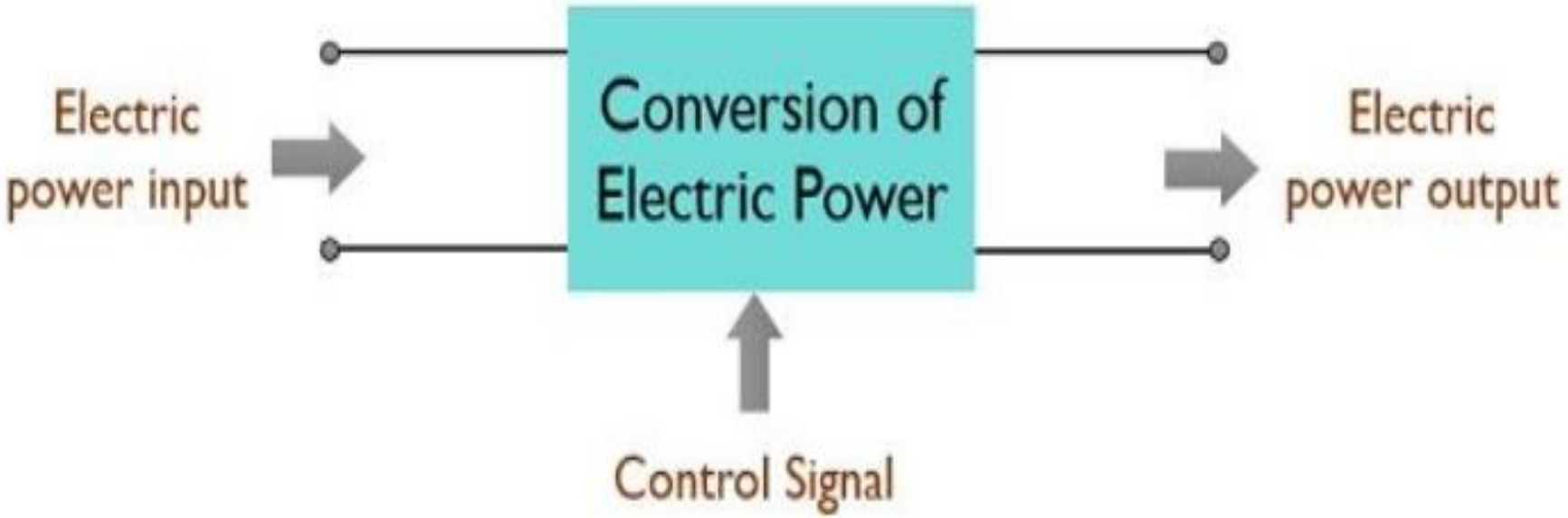
POWER ELECTRONICS- Advantages

- **High efficiency** due to low loss in semiconductor devices.
- **High reliability** of power electronic converter systems.
- **Long life & less maintenance** due to absence of any moving parts.
- **Fast response** as compared to electromechanical converter systems.
- **Small size, less weight**, so less floor space, low installation cost.
- Mass production of semiconductor devices result in **low cost of the converter equipment.**

POWER ELECTRONICS- Disadvantages

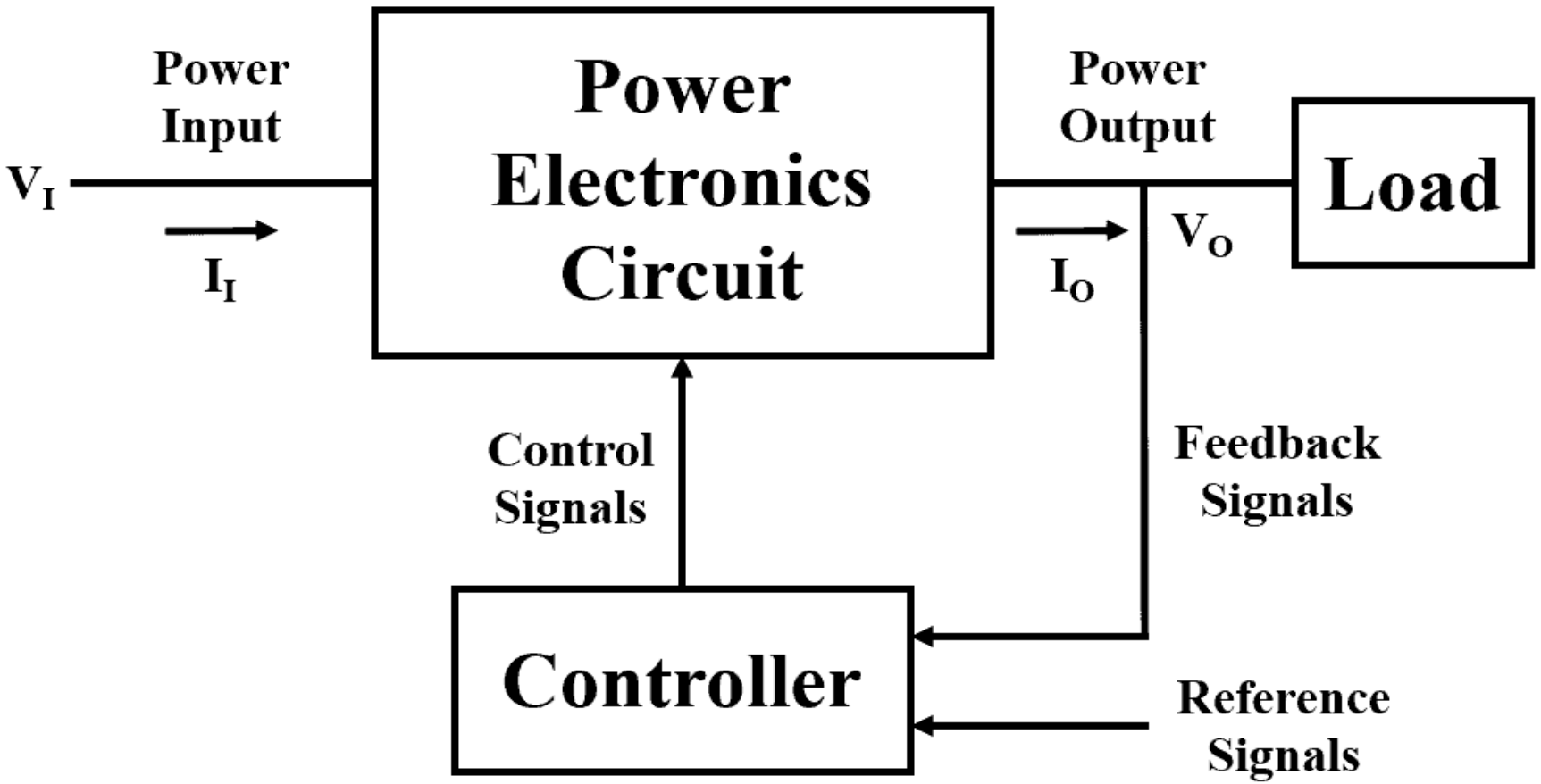
- Power electronic converters have a **tendency to generate harmonics** in the supply system & in the load circuit.
- AC to DC and AC to AC converters operate at low input power factor. So some **measures have to be adopted to improve power factor**.
- Power electronic controllers have **low overload capacity**. So these controllers must be rated for taking momentary overloads. This may increase the cost of power electronic controllers.
- **Regeneration of power is difficult** in power electronic converter systems.

Power Electronic System



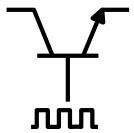
Representation of Electric Power Conversion

Power Electronic System (Contd…)



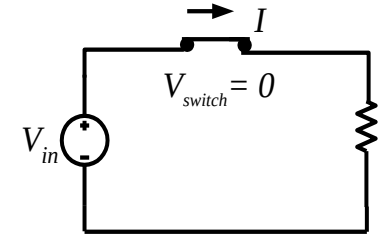
Power Electronic Switches

- **Power switches**: work-horses of PE systems.



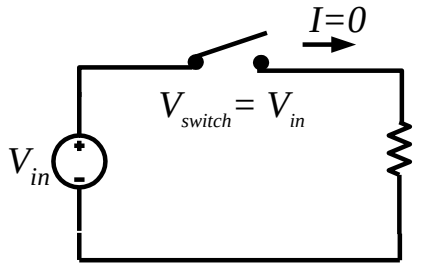
POWER SWITCH

- Operates in two states:
  - Fully **on**. i.e. switch closed.
  - **Conducting** state



SWITCH ON (fully closed)

  - Fully **off** , i.e. switch opened.
  - **Blocking** state



SWITCH OFF (fully opened)

POWER SEMICONDUCTOR DEVICES- History

- **Mercury-arc rectifiers**
- Robust and compact **-Silicon Controlled Rectifier (SCR)** replaced mercury-arc rectifiers
- Then came **bipolar transistors** which handle few tens of amperes and block a few hundred volts **-** superior in its turn-off capability.
- **Power MOSFET** – integration of modern integrated circuit and discrete power semiconductor manufacturing technologies **-** higher operating frequencies upto a few MHz **-** high voltage MOSFETs are not commercially viable
- **Gate-turn-off thyristor (GTO)** – was developed as an improvement on SCR regarding its turn-off capability**-**application restricted to inverters rated above a few hundred kVA
- **Insulated Gate Bipolar Transistor (IGBT)** – basically a MOSFET driven Bipolar device – available upto 4 kV and 4 kA**-** switching frequency is 25 kHz

POWER SEMICONDUCTOR DEVICES

1. Power Diodes
2. Power transistors
3. Power MOSFETs
4. IGBTs
5. Thyristors

Based on turn-on and turn-off chara and gate signal requirements devices are classified as:-

- **Diodes** – uncontrolled rectifying devices – on and off states controlled by power supply
- **Thyristors** (SCRs)– controlled turn-on by a gate signal
- **Controllable switches** – turned-on and turned-off by the application of control signals – BJT, MOSFET, IGBT, GTO, SITH, MCT, SIT

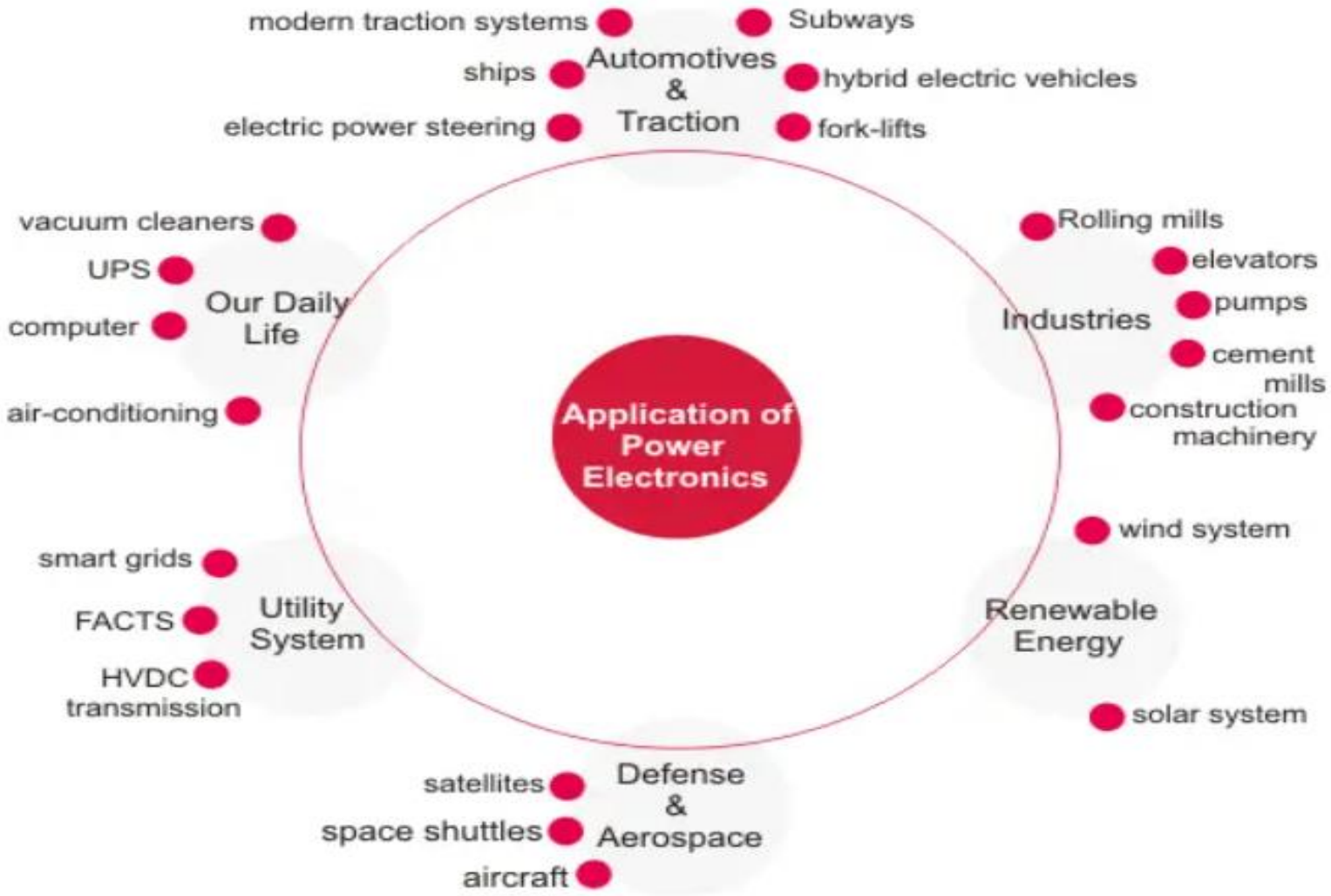
POWER SEMICONDUCTOR DEVICES (Contd...)

- **SCR**(Silicon Controlled Rectifier), **GTO**(Gate Turn-Off Thyristor), **MCT** (MOSFET Controlled Thyristor) and **SIT**(Static Induction Transistors) – require pulse signal for turn-on and once ON the pulse can be removed.
- **BJT** (Bipolar Junction Transistor), **MOSFET** (Metal Oxide Semiconductor Field Effect Transistor), **IGBT** (Insulated Gate Bipolar Transistors) and **SIT** – require continuous signal for keeping them in the turn-on condition
- **BJT, MOSFET, IGBT** and **MCT** – withstand unipolar voltages
- **Thyristors** and **GTOs** – can withstand bipolar voltages
- **TRIAC** and **RCT** (Reverse Conducting Thyristor) – possess bidirectional current capability
- **Diodes, SCR, BJT, MOSFET, IGBT, GTO, SIT, MCT** – unidirectional current devices

POWER SEMICONDUCTOR DEVICES - Ratings

Devices	Voltage and Current (upto)	Frequency
Power Diodes	3000 V, 3500 A	1 kHz
Power Transistors	1200 V, 400 A	10 kHz
Power MOSFETs	1000 V, 50 A	100 kHz
IGBTs	1200 V, 400 A	20 kHz
GTOs	4000 V, 3000 A	10 kHz
MCTs	600 V, 60 A	20 kHz
TRIAC	1200 V, 300 A	0.4 kHz
Static Induction Transistors (SITs)	1200 V, 300 A	100 kHz
Static Induction Thyristors (SITHs)	4000 V, 2000 A	20 kHz

Power Electronics-Applications (Contd…)



Power Electronics-Applications (Contd…)

- **Commercial applications:-** Heating and Ventilating Systems, Air Conditioners, Central Refrigeration, Lighting, Computers and Office equipments, Uninterruptible Power Supplies (UPS), Elevators, and Emergency Lamps
- **Domestic applications:-** Cooking Equipments, Food mixers, Vacuum cleaners, Dryers, Fans
- **Industrial applications:-** Pumps, Compressors, Blowers and Fans, Machine tools, Arc furnaces, Induction furnaces, Lighting control circuits, Industrial lasers, Induction heating, Welding equipments
- **Telecommunication:-** Battery chargers, power supplies (DC and UPS), mobile cell phone battery chargers

Power Electronics-Applications (Contd…)

- **Aerospace applications:-** Space shuttle power supply systems, satellite power systems, aircraft power systems.
- **Transportation:-** Traction control of electric vehicles, battery chargers for electric vehicles, electric locomotives, street cars, trolley buses, automobile electronics including engine controls.
- **Utility systems** High voltage DC transmission (HVDC), static VAR compensation (SVC), Alternative energy sources (wind, photovoltaic), fuel cells, energy storage systems, induced draft fans and boiler feed water pumps

Power Electronics *vs* linear Electronics

- In Power Electronics all devices are operated in the switching mode – either 'FULLY-ON' or 'FULLY-OFF' states.
- Power electronics focuses on efficiency and power conversion, often resulting in lower heat dissipation, and losses
- Linear electronics uses components in their "active" or "linear" region to process signals, typically for amplification or signal conditioning.
- Linear electronics focuses on signal fidelity with higher efficiency losses and power consumption.

Power Electronics *vs* Signal Electronics

Basis of comparison	**Signal Electronics**	**Power Electronics**
Definition	the field of engineering and physics that deals with the study of behavior, movement and effects of electrons and with the solid state devices.	the branch of electrical engineering which deals with the **behavior and operation of high voltage and current devices used to control and deliver electrical power.**
Voltage and current rating	works with low voltage and low current ratings.	works with high voltage and high current ratings.
Noise	noise and distortion are high.	**low noise and distortion**.
Electrical safety hazards	there is no or very low electrical safety hazards.	involves **high electrical safety hazards.**

Power Electronics *vs* Signal Electronics (Contd…)

Basis of comparison	**Signal Electronics**	**Power Electronics**
Purpose	to capture and analyze the data of a system.	to **operate a system efficiently**.
Power losses	low power losses	**the power losses are relatively high**
Devices	Some examples using electronic devices are computers, smartphones, FM radios, TVs, watches, diodes, signal transistors, etc.	Some examples using practical power electronics devices are SMPS, battery chargers, VFDs, UPS, Inverters, etc.
Applications	used in amplification, rectification, multiplexing, encoding, wireless transmission of signals, data processing, etc.	used in **control, rectification, inversion, conversion, etc. of high electrical power**.

Power Electronics - benefits

- Electric power is scarce, so prime importance to deliver the power to the load with minimum losses.
- Advancements in power semiconductor research has helped in achieving this.

The **benefits** are:-

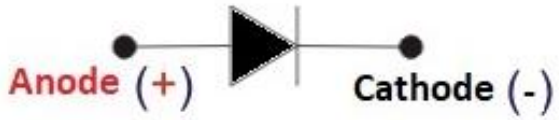
- High power density power supplies
- Improved efficiency of up to 99% in power conversion
- Noise-sensitive applications such as in medical devices
- Power can be made available in the desired form and level as per specifications
- Increased clean power generation using power electronic interfaces to connect the sources to grid
- Wireless power transfer

POWER SEMICONDUCTOR DEVICES

1. Power Diodes



Power Diodes

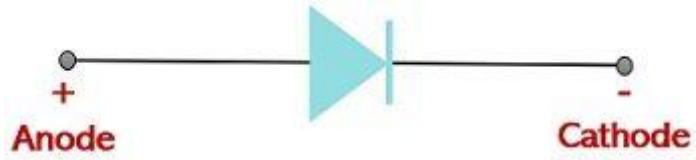


Power Electronic Devices - features

Power Diodes:-

3 types –

- Line frequency diodes
- Fast recovery diodes
- Schottky diodes



Symbol of Power Diode

Line frequency diodes:-

- Highest voltage & current – 5 kV & 5 kA
- Excellent over current rating
- Excellent voltage withstand capability

Power Electronic Devices – features (Contd…)

Fast recovery diodes:-

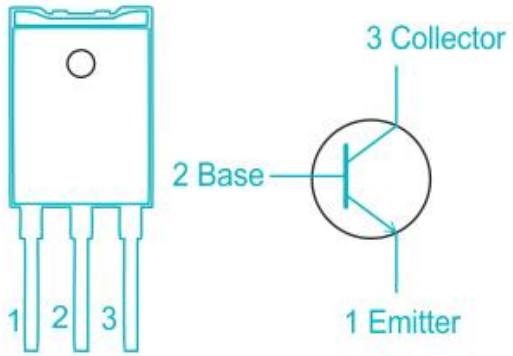
- Available at high power
- Used in association with fast controlled devices
- Used in UPS, traction, induction heating etc.

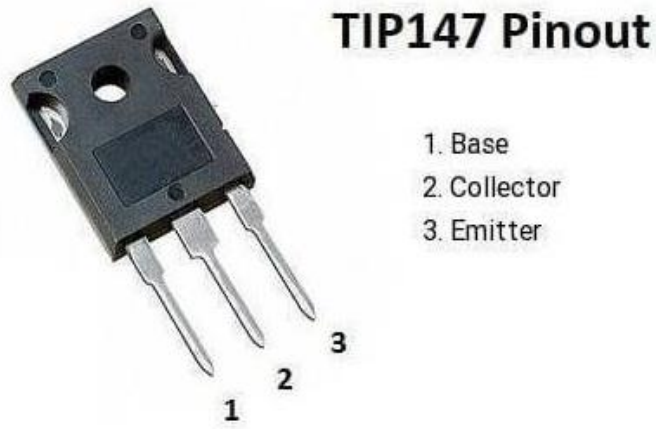
Schottky diodes:-

- Fastest diodes
- Majority carrier device
- Voltage rating only upto a 100 V
- High current rating

POWER SEMICONDUCTOR DEVICES (Contd....)

2. **Power Transistors**

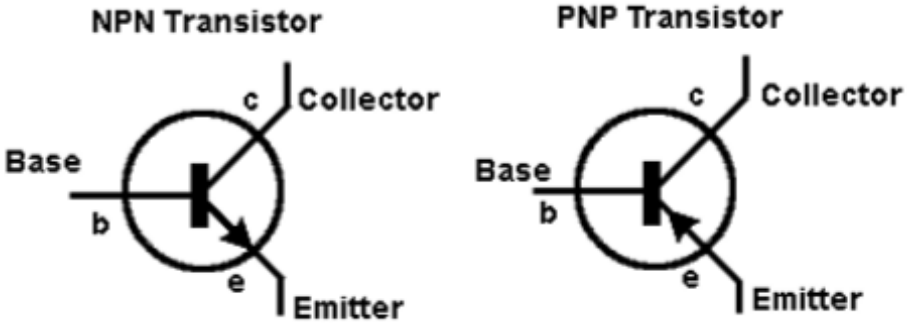




Power Electronic Devices – features (Contd…)

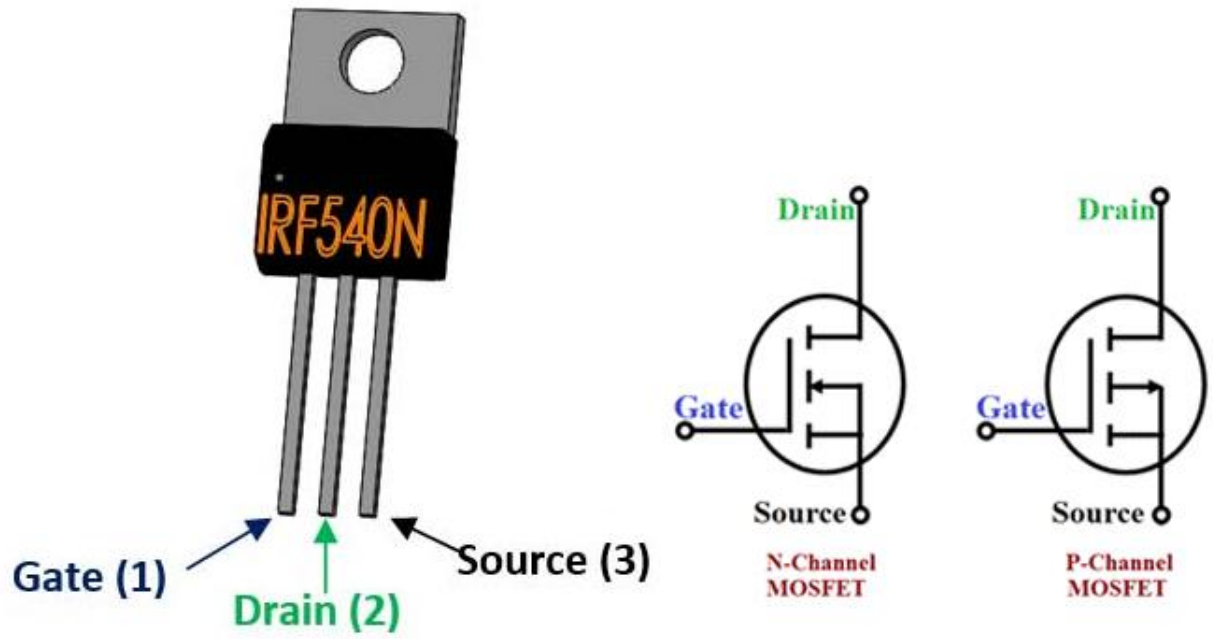
Power Bipolar transistor:-

- Developed to handle few tens of amperes & block few hundred volts
- High operating frequency is possible
- Not capable of blocking reverse voltage, so application is limited
- Superior the turn-off capability
- Preferred in forced commutated converters
- But gain is somewhat less



POWER SEMICONDUCTOR DEVICES (Contd....)

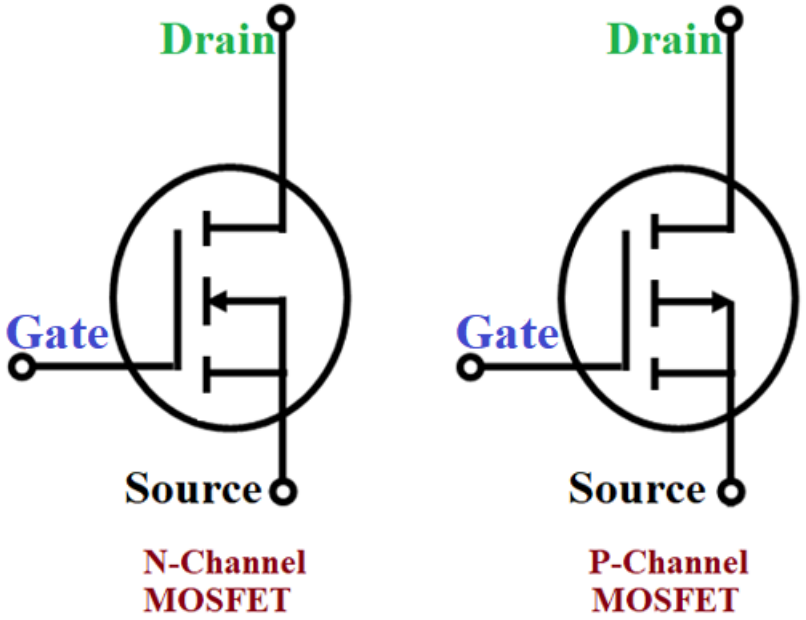
3. **MOSFET-** Metal Oxide Semiconductor Field Effect



Power Electronic Devices – features (Contd…)

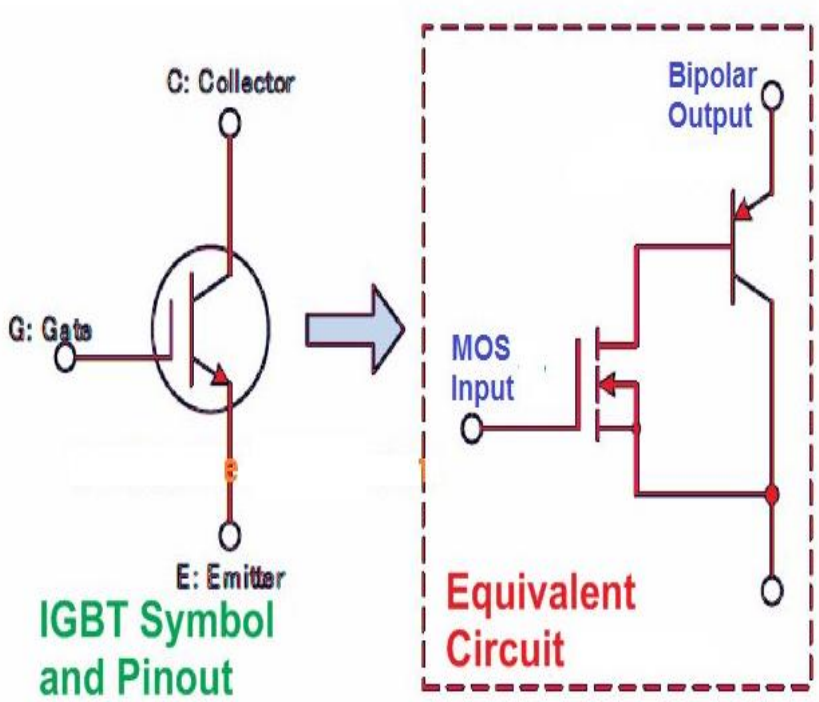
Power MOSFET:-

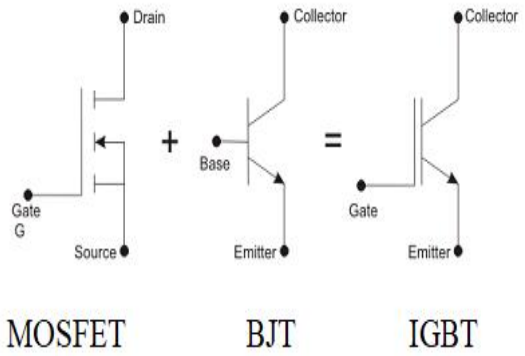
- High gain
- Ease of paralleling
- High operating frequency
- Voltage fed device
- Majority carrier device
- Needs only a very low current to switch on
- Delivers high current of more than 50 A
- High voltage MOSFETs are not commercially viable
- Preferred where high switching frequencies are desired but operating voltages are low

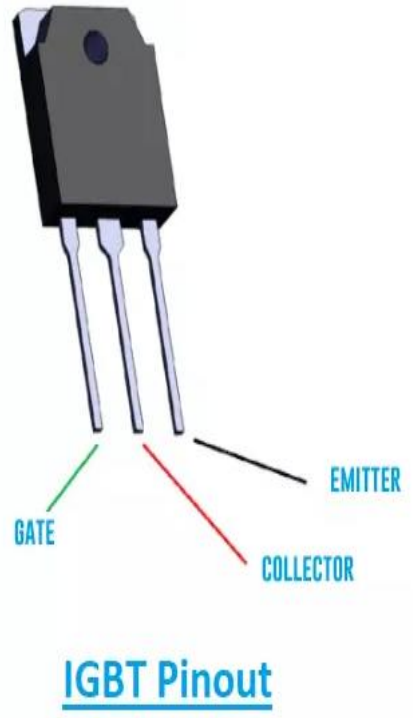


POWER SEMICONDUCTOR DEVICES (Contd....)

4. IGBT- Insulated Gate Bipolar Transistor



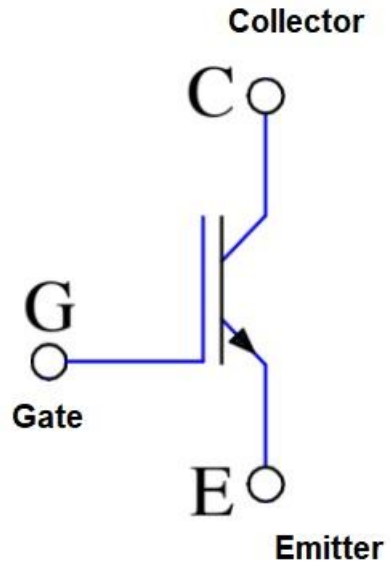




Power Electronic Devices – features (Contd…)

IGBT:-

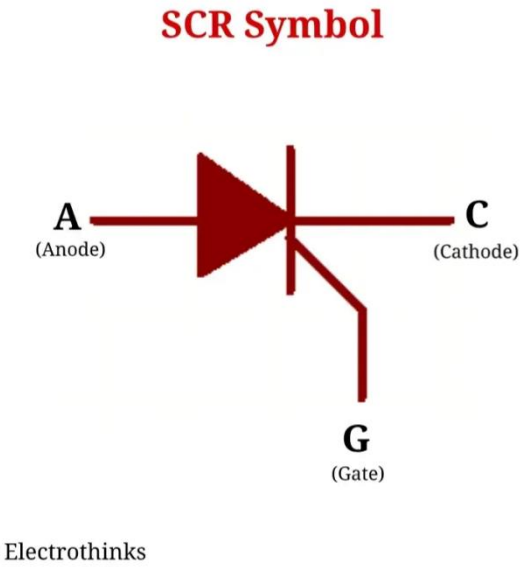
- MOSFET + Bipolar
- Voltage controlled device
- unidirectional
- Available upto 1200 V
- Switching frequency is approx. 25 kHz
- Ease of connection
- Requires only a small voltage to maintain conduction through the device
- Does not require protection circuits
- Can be connected in parallel without difficulty
- Replaced BJT in almost all applications



THYRISTOR FAMILY– features

SCR:-

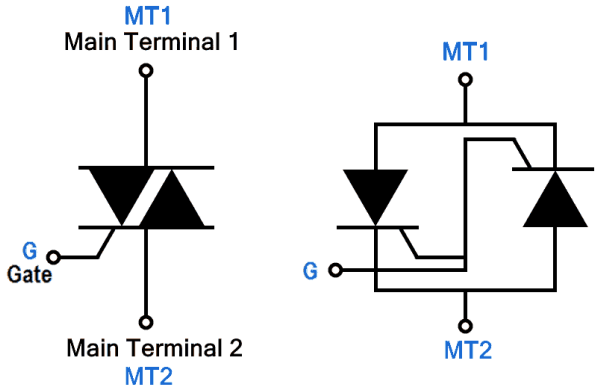
- Most popular in the thyristor family
- **Unidirectional** device
- Turned-on by the application of a gate signal
- Voltage rating of forward and reverse blocking are almost equal
- 2 types – converter grade (phase controlled thyristor) and inverter grade
- Converter grade – main element of PE- available at voltage ratings above 5 kV and 5 kA
- Inverter grade – called fast grade SCRs



THYRISTOR FAMILY– features (Contd…)

TRIAC:-

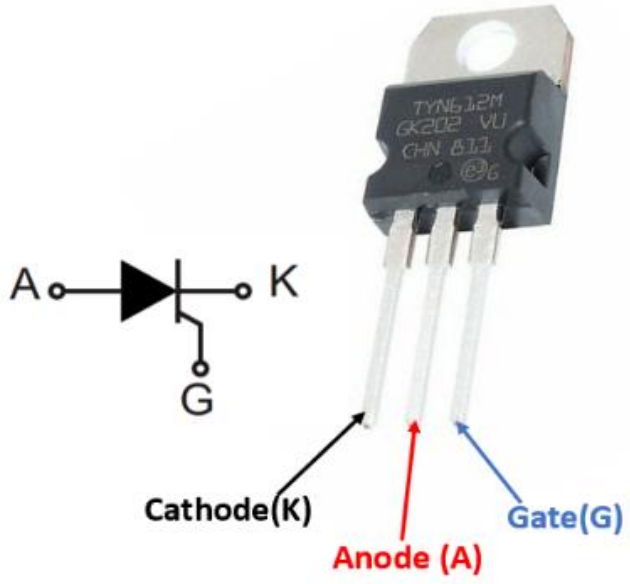
- TRIODE for alternating current(**TRI**ode + **AC**)
- **Bidirectional**
- 3 terminal device
- Controls the flow of current
- Similar to 2 thyristors connected back to back
- Regarded as DIAC with an extra gate terminal



TRIAC Symbol

POWER SEMICONDUCTOR DEVICES (Contd....)

5. SCR- Silicon Controlled Rectifier or Thyristors









THYRISTOR FAMILY– features (Contd…)

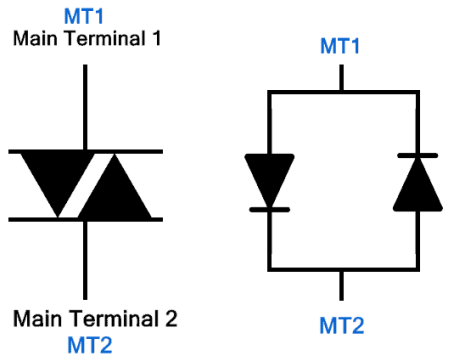
TRIAC:- (Contd...)

- Used in applications which does not require very high power
- Used in light dimmers, small electric fans etc.
- Facilitates switching operation in function in AC electrical components & systems.
- Not used in high power applications because it causes EMI.

THYRISTOR FAMILY– features (Contd…)

DIAC:-

- DIODE for alternating current(**DI**ode + **AC**)
- **Bidirectional depending on the voltage across its terminals**
- 2 terminal device
- Conducts when the voltage across the terminals is greater than the breakdown voltage
- Similar to 2 diodes connected back to back
- Used for triggering of TRIACs, speed control of universal motors

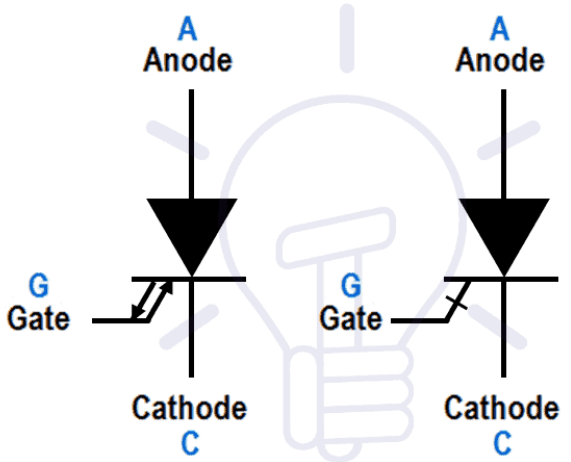


DIAC Symbol

THYRISTOR FAMILY– features (Contd…)

GTO:-

- High cost
- High switching speed than SCR
- Can be turned-on by a short pulse of gate signal and turned-off by a reverse gate signal.
- Used in inverters rated above a few 100 kVA

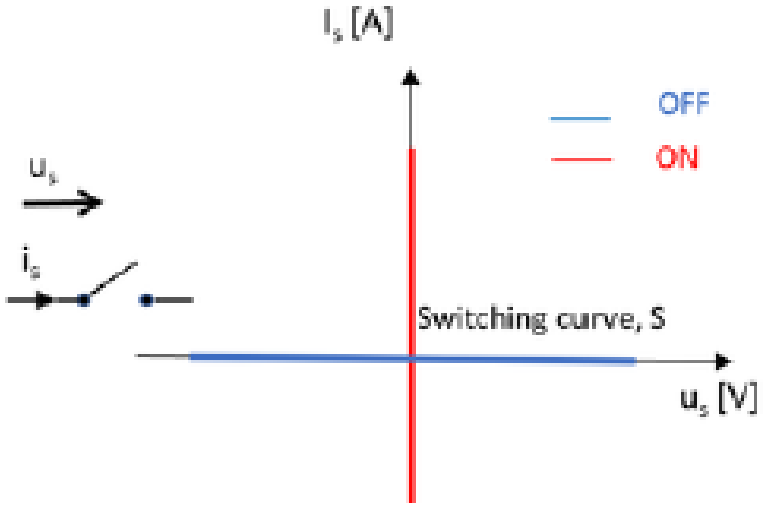


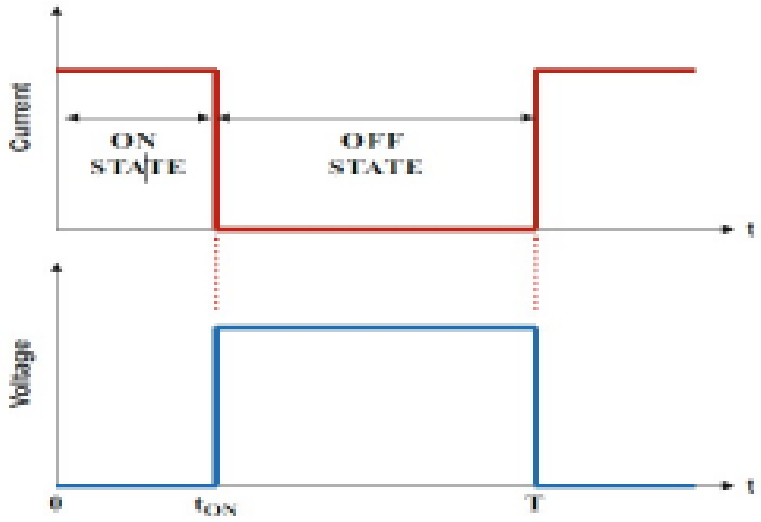
Gate Turn-Off (GTO) Thyristor Symbol

POWER SEMICONDUCTOR DEVICES - Ratings

Devices	Voltage and Current (upto)	Frequency
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TRIAC	1200 V, 300 A	0.4 kHz
Static Induction Transistors (SITs)	1200 V, 300 A	100 kHz
Static Induction Thyristors (SITHs)	4000 V, 2000 A	20 kHz

IDEAL SWITCH



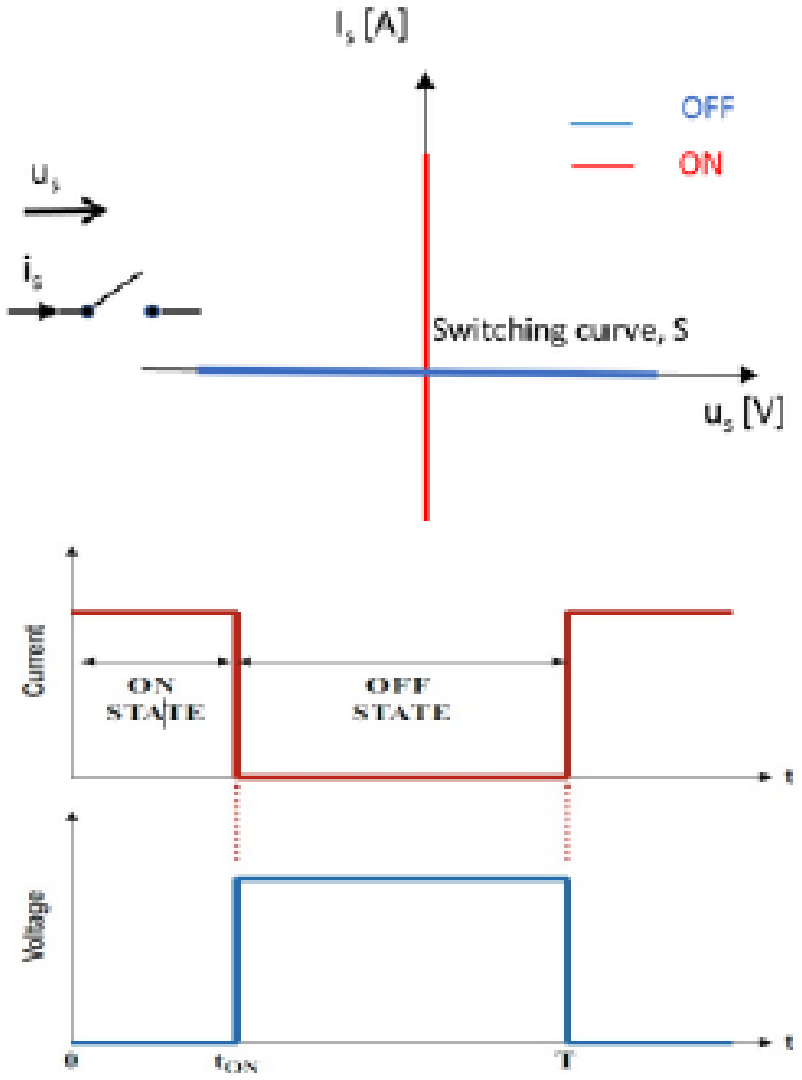


- Off-state Infinite resistance;
- Zero leakage current.

- On-state Zero voltage drop across the switch;
- Infinite current handling capability.

- Switching speed Instantaneous (zero turn-on and turn-off times)
- Power loss Zero, as either voltage or current is zero at all times.

IDEAL SWITCH



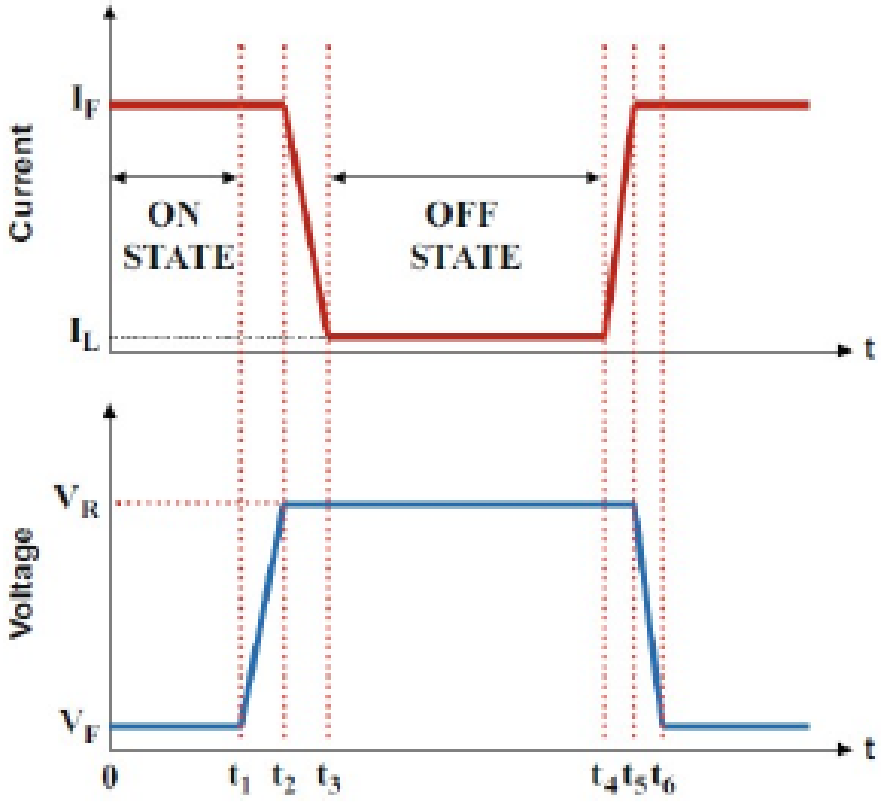
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- Power loss Zero, as either voltage or current is zero at all times.

IDEAL SWITCH	
On-state	Zero voltage drop across the switch; infinite current handling capability.
Off-state	Infinite resistance; zero leakage current.
Switching speed	Instantaneous (zero turn-on and turn-off times)
Power loss	Zero, as either voltage or current is zero at all times.

PRACTICAL SWITCH



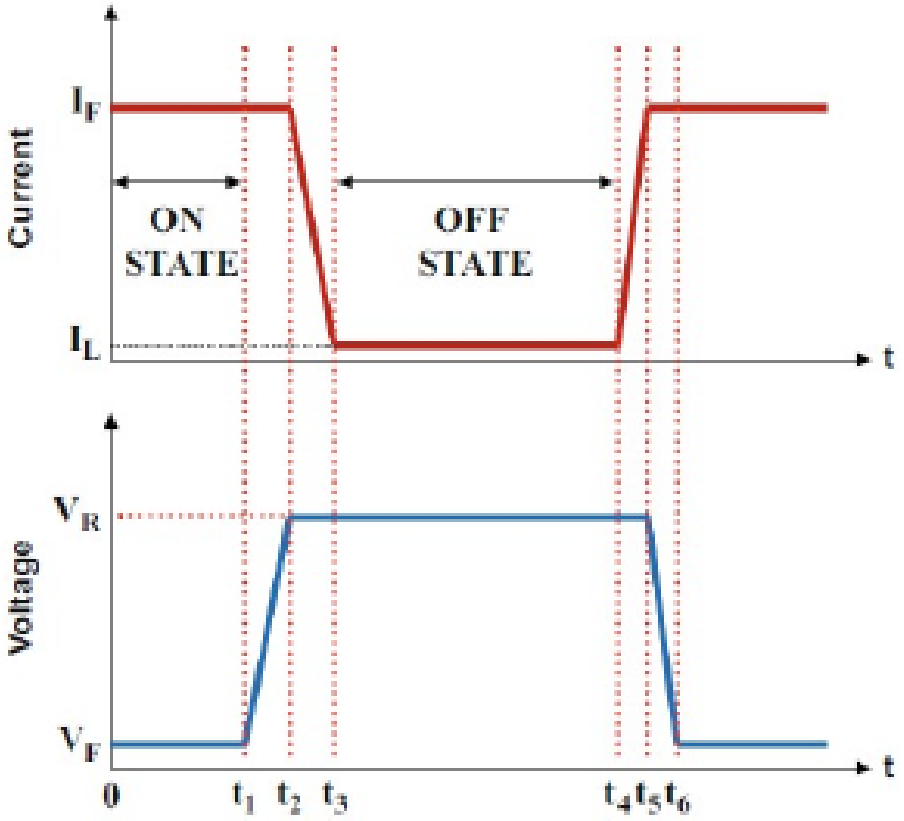
Off-state:

- Finite off-state resistance and
- a non-zero leakage current which leads to power loss even when the switch is off.

On-state:

- Non-zero voltage drop and
- a finite on-state resistance (Ron) which leads to conduction loss.

PRACTICAL SWITCH



Switching speed:

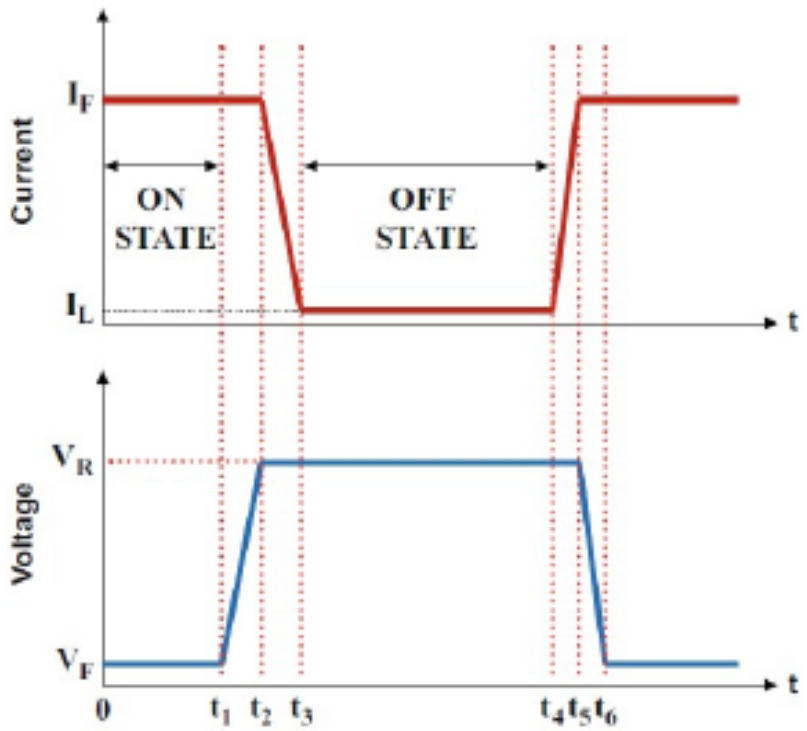
Finite turn-on and turn-off times, which limits the maximum switching frequency.

Power loss:

Includes conduction loss (due to on-state resistance) and

switching loss (due to finite switching times when both voltage and current are momentarily non-zero).

PRACTICAL SWITCH



Other limitations:

Limited voltage and current ratings, and

limited rates of change for current (di/dt) and voltage (dv/dt)

Power losses in a Switch

